



AI & ROBOTICS BASED INTELLISHOP

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Objective

1. Automation of Customer Assistance
2. Advanced Object Detection
3. Seamless User Interaction
4. Ensuring Secure and Dependable Operations

Technology Used

Hardware used:

Arduino, NodeMCU, Esp32s camera, L293d motor shield

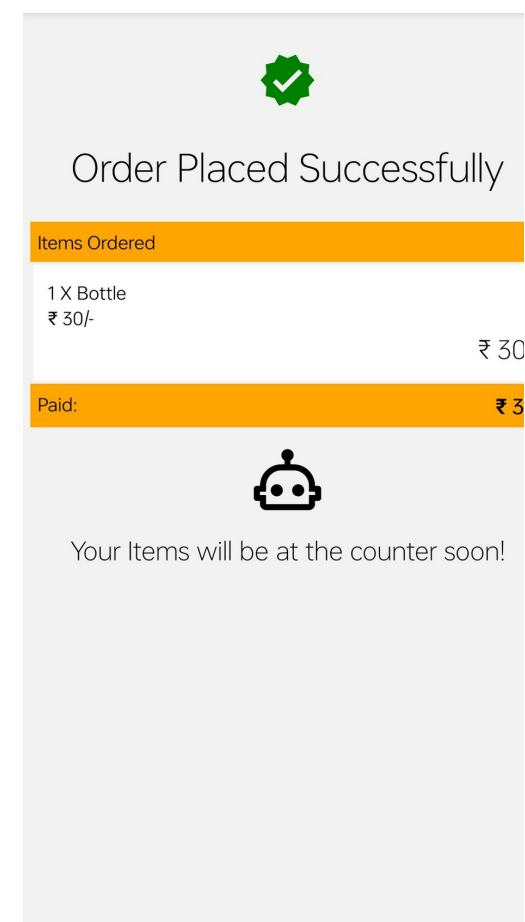
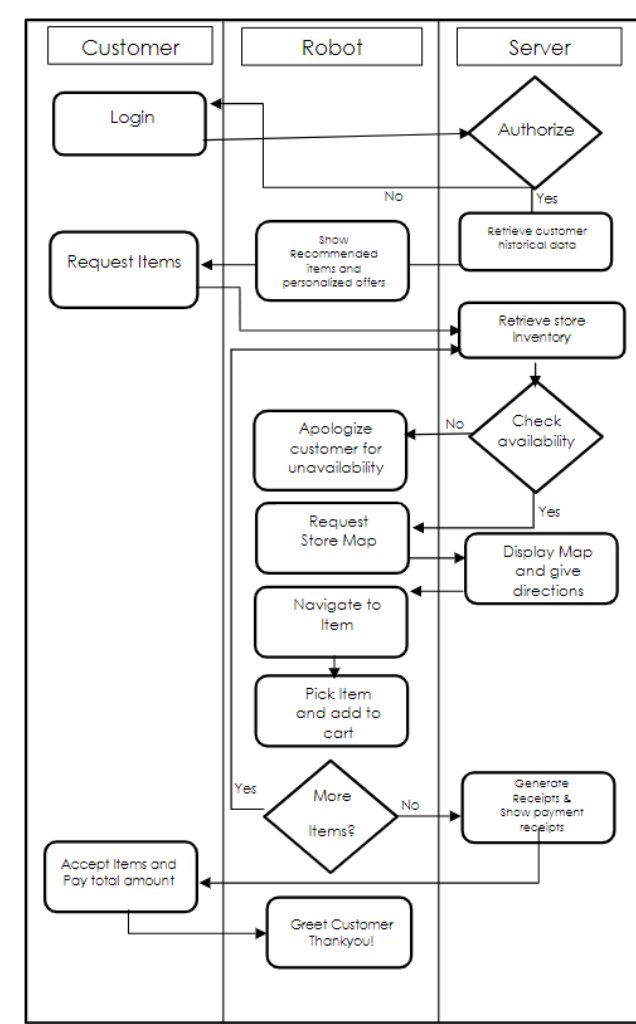
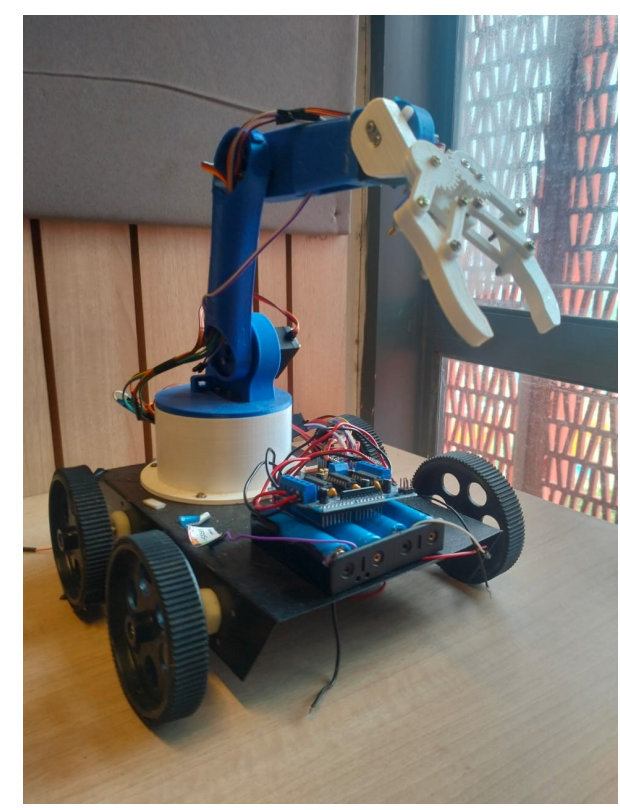
Deep Learning model used:

Yolov3

Frontend and Backend technologies used:

HTML5, CSS, Vannila Javascript, React Native and Firebase

Arduino Programming



Novelty Of Work

- Real Time Adaptive control Mechanism
- Potential for scalability and adaptability
- Integration of advanced computer vision and machine learning techniques with a meticulously designed robotic arm

Future Scope and Outcomes

Future Scope

- Scalability and adaptability
- Navigation Improvisation

Outcomes

- Successful Integration of AI and Robotics
- Operational Efficiency and Cost Savings

Test Cases

- Test Case 1 : Bottle Detection
- Test Case 2 : New User Sign up
- Test Case 3 : Bottle picked up by the robot
- Test Case 4 : Bottle detected in a not so clear background